

ANDRE GLEGG

Senior Software Engineer | TypeScript | React | Node.js | 3D/WebGL | AI Agents | Distributed Systems
Oslo, Norway | +47 930 15 729 | andreglegg@me.com
andreglegg.no | github.com/andreglegg | linkedin.com/in/andre-glegg-060a3164 | medium.com/@andreglegg

PROFESSIONAL SUMMARY

Senior software engineer with 15+ years building production web, 3D, realtime and full-stack systems. Former team lead at Fjong and senior engineer at Aize, with deep hands-on experience in TypeScript, React, Angular, Node.js and Three.js/WebGL. Owns systems from architecture through performance, testing, CI/CD and production operations. Recent independent work includes a benchmarked local-LLM coding agent, distributed inference in Rust, procedural 3D tooling and released cross-platform games.

TECHNICAL EXPERTISE

Languages: TypeScript, JavaScript, C#, Rust, HTML/CSS
Frontend & 3D: React, Angular, Three.js, WebGL, RxJS, Zustand, Vite, Godot, procedural geometry, GLB/OBJ
Backend & Data: Node.js, Express, WebSockets, REST, GraphQL, PostgreSQL, Redis, Elasticsearch, Supabase
AI & Developer Tools: LLM agents, local LLMs, MCP, TypeScript Compiler API, evaluation/benchmarking, distributed inference, QUIC/iroh
Testing & Delivery: Jest, Vitest, React Testing Library, Playwright, Docker, CI/CD, Azure DevOps, Fly.io, Firebase
Leadership: Technical architecture, mentoring, code review, hiring, delivery ownership, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Independent Software Engineer & Product Builder | Personal & Open-Source Projects

Norway | Jul 2024 - Present

- Design, build, ship and operate software across AI developer tooling, distributed systems, realtime multiplayer, procedural 3D and mobile products.
- Built Forge, Cognara and treegen; shipped LastCoil and Endless Descent on iOS and Android; maintain public/open-source code and hosted developer tooling.
- Own architecture, implementation, networking, persistence, testing, benchmarking, observability, deployment and release automation, with performance and soak tests used as engineering gates.

Senior Software Engineer | Aize AS

Oslo, Norway | Aug 2021 - Jun 2024

- Built and maintained browser-based 3D digital-twin viewers and engineering-document tools for navigating large, complex industrial datasets.
- Implemented TypeScript/JavaScript features across React/Angular, Three.js/WebGL visualization, model navigation, annotations, document workflows and client-side state/data processing.
- Improved rendering, geometry-loading and interaction performance, and strengthened automated testing and reliability in complex data-heavy user flows.
- Owned features end to end, collaborated with product/design and engineering peers, reviewed code and contributed to architecture and maintainability across the product.

Full-Stack Developer / Team Lead | Fjong Norge AS

Oslo, Norway | Jun 2018 - Aug 2021

- Led a four-person development team and owned delivery for a fashion-rental platform spanning rental flows, inventory, subscriptions, internal tools, releases and cloud operations.
- Built full-stack systems with JavaScript/TypeScript, Node.js/Express, .NET/C#, databases and Azure infrastructure.
- Designed Elasticsearch search/discovery with custom analyzers, synonym and brand handling, availability-aware filtering, relevance scoring, faceted navigation and type-ahead suggestions.
- Improved CI/CD and operational automation while mentoring developers, reviewing code and coordinating technical priorities.

SELECTED ENGINEERING PROJECTS

Forge | AI Coding Agent | TypeScript | Local LLMs

Open-source public alpha | 2026 | github.com/andreglegg/forge

- Built a local-first coding-agent CLI optimized for local and small language models, with Git-aware repository indexing, scoped search/read tools, TypeScript symbol/caller analysis, transactional edits, command execution and verification gates.
- Engineered permission modes, isolated Git worktrees, patch-risk scanning, crash-tolerant sessions/undo, trace replay, Docker/Podman execution, MCP client support and GitHub issue-to-draft-PR workflows.
- Benchmark-gated development achieved 60.0% (135/225) on Aider Polyglot, with pinned 42-case screens reaching 28/42 and 27/42 across two Forge runs.

Cognara | Distributed AI | Rust | QUIC/iroh

Private research project | 2026

- Built a peer-to-peer distributed-inference prototype that pools compute and memory across ordinary machines so model partitions can execute across multiple nodes.
- Implemented pipeline-sharded inference over QUIC/iroh and validated multi-node operation with long-running soak tests and 16K-context workloads across a three-node cluster.

LastCoil | Realtime Multiplayer | Three.js | Node.js

iOS & Android | 2025 - Present

- Built and shipped a realtime multiplayer mobile game using Three.js, TypeScript, Node.js, WebSockets, PostgreSQL and Redis, packaged with Capacitor and deployed on Fly.io.
- Implemented bots, cosmetics, leaderboards, persistence, notifications and Firebase integrations alongside type checks, Vitest/Playwright coverage, smoke tests and capacity/performance gates.

treegen | Procedural 3D | Three.js | MCP

Public developer tool | 2026 | github.com/andreglegg/treegen

- Created a deterministic low-poly tree generator and public MCP service that exports game-ready GLB/OBJ assets from the browser, Node.js and AI coding clients.
- Built species-aware procedural skeleton/mesh generation, lifecycle controls and LOD export, with automated visual and structural regression checks.

TECHNICAL WRITING

Medium - AI agents, distributed systems and software engineering. medium.com/@andreglegg

Tool Design Beats Clever Prompts - tool architecture, observability, validation and reliable LLM software.

The Agent Loop Is Not a Workflow - agent state, verification, control loops and termination.

Home Computers Are Becoming Tiny AI Datacenters - distributed inference and consumer compute.

EARLIER EXPERIENCE

Programming Instructor, Bitcamp / StartupLab, Oslo | Jan 2018 - Taught programming fundamentals to students aged 7-16 at Oslo International School.

Freelance Programmer, Jamaica | 2011 - 2017 - Built websites and mobile applications for iOS and Android for independent clients.

Web & Mobile Application Developer, Irie FM / Zip FM, Jamaica | Oct 2008 - Nov 2011 - Built websites, digital media systems, Linux servers and Shoutcast streaming infrastructure.

Additional Norway experience | 2017 - Frontliner, Greenpeace Norway.

PROFESSIONAL DEVELOPMENT & LANGUAGES

Certificates: Android Apps, University of Illinois (Coursera), 2014; Web Page Design & Development, Vector Technology Institute, Jamaica, 2013.

Languages: English: Fluent | Norwegian: Basic / actively improving